DBMS Question Paper – Answers

Section A – Short Answer Questions

1. Define DBMS and explain its advantages over the file system. (3 Marks)

A **Database Management System (DBMS)** is software used to create, manage, and organize databases efficiently. It allows users to store, retrieve, update, and manage data in a structured manner.

Advantages over File System:

- **Reduced Data Redundancy:** Avoids duplicate data storage.
- **Data Security:** Provides authorized access to data.
- **Data Integrity:** Maintains accuracy and consistency of data.
- **Data Sharing:** Multiple users can access data simultaneously.
- **Backup and Recovery:** Easier recovery of data during failures.

---

2. Differentiate between Primary Key and Foreign Key with examples. (3 Marks)

Primary Key	Foreign Key
Uniquely identifies each record in a table	Establishes relationship between two tables
Cannot contain NULL values	Can contain NULL values
Only one primary key per table	Multiple foreign keys possible

Example:

- **Primary Key:** `StudentID` in Student table.
- **Foreign Key:** `StudentID` in Marks table referencing Student table.

3. Explain the concept of Normalization and its importance in database design. (3 Marks)

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

Importance:

- Eliminates duplicate data.
- Improves consistency.
- Simplifies database maintenance.
- Reduces anomalies during insertion, deletion, and updates.

Common normal forms include 1NF, 2NF, and 3NF.

---

4. What is an ER Model? Explain any two components of an ER diagram. (3 Marks)

An **Entity Relationship (ER) Model** is a conceptual design used to represent the structure of a database visually.

Components:

1. **Entity:** Represents a real-world object or table.
   Example: Student, Employee.
2. **Attribute:** Describes properties of an entity.
   Example: StudentID, Name, Age.

---

5. Define Transaction in DBMS. Explain the ACID properties. (3 Marks)

A **Transaction** is a sequence of database operations performed as a single logical unit of work.

ACID Properties:

- **Atomicity:** Transaction is completed fully or not at all.
- **Consistency:** Database remains valid before and after transaction.
- **Isolation:** Transactions execute independently.
- **Durability:** Changes remain permanently after commit.

Section B – Medium Answer Questions

6. Explain the three-schema architecture of DBMS. (5 Marks)

The **three-schema architecture** separates the database into three levels:

1. **External Level (View Level):**
    - User-specific views of data.
    - Different users see different parts of database.
2. **Conceptual Level (Logical Level):**
    - Describes overall database structure.
    - Defines relationships, entities, and constraints.
3. **Internal Level (Physical Level):**
    - Describes how data is physically stored in memory.

Advantages:

- Data abstraction.
- Data independence.
- Improved security and maintenance.

---

7. Describe the different types of joins in SQL with examples. (5 Marks)

SQL joins are used to combine data from multiple tables.

Types of Joins:

1. INNER JOIN

Returns matching records from both tables.

SELECT *

FROM Student

INNER JOIN Marks

ON Student.StudentID = Marks.StudentID;

2. LEFT JOIN

Returns all records from left table and matching records from right table.

SELECT *

FROM Student

LEFT JOIN Marks

ON Student.StudentID = Marks.StudentID;

3. RIGHT JOIN

Returns all records from right table and matching records from left table.

SELECT *

FROM Student

RIGHT JOIN Marks

ON Student.StudentID = Marks.StudentID;

4. FULL OUTER JOIN

Returns all records when there is a match in either table.

SELECT *

FROM Student

FULL OUTER JOIN Marks

ON Student.StudentID = Marks.StudentID;

Classic SQL joins be like: “you complete me.” 😭

---

8. Explain Functional Dependency and its role in normalization. (5 Marks)

A **Functional Dependency (FD)** occurs when one attribute uniquely determines another attribute.

Example:

If `StudentID → StudentName`, then StudentID determines StudentName.

Role in Normalization:

- Helps identify redundant data.
- Used to achieve higher normal forms.
- Ensures proper database structure.
- Prevents update anomalies.

Functional dependency is essential for converting tables into 2NF, 3NF, and BCNF.

---

9. Discuss the differences between DDL, DML, and DCL commands with examples. (5 Marks)

Type	Description	Examples
DDL (Data Definition Language)	Defines database structure	CREATE, ALTER, DROP
DML (Data Manipulation Language)	Manipulates data	INSERT, UPDATE, DELETE
DCL (Data Control Language)	Controls access permissions	GRANT, REVOKE

Examples:

DDL

CREATE TABLE Student (

StudentID INT,

```
    Name VARCHAR(50)
);
```

DML

```
INSERT INTO Student
VALUES (101, 'Ravi');
```

DCL

```
GRANT SELECT ON Student TO user1;
```

---

10. Explain indexing in DBMS and its advantages. (5 Marks)

An **Index** is a database object used to improve the speed of data retrieval operations.

It works similarly to an index in a book — instead of reading every page, DBMS quickly finds the required data.

Advantages:

- Faster data retrieval.
- Improves query performance.
- Reduces search time.
- Efficient sorting and filtering.

Disadvantages:

- Requires extra storage space.
- Slows down insert and update operations slightly.

Example:

```
CREATE INDEX idx_name
ON Student(Name);
```

Section C – Technical Query Question

Given Table:

StudentID	Name	Department	Marks
101	Ravi	CSE	85
102	Anu	ECE	78
103	Kiran	CSE	92
104	Meena	EEE	88

11(a) Display the names of students who belong to the CSE department. (3 Marks)

```
SELECT Name

FROM Student

WHERE Department = 'CSE';
```

Output:

Name

Ravi

Kiran

11(b) Find the average marks of all students. (3 Marks)

```
SELECT AVG(Marks) AS Average_Marks

FROM Student;
```

Output:

Average_Marks
85.75

11(c) Display the student details in descending order of marks. (4 Marks)

```
SELECT *

FROM Student

ORDER BY Marks DESC;
```

Output:

StudentID	Name	Department	Marks

103	Kiran	CSE	92
104	Meena	EEE	88
101	Ravi	CSE	85
102	Anu	ECE	78